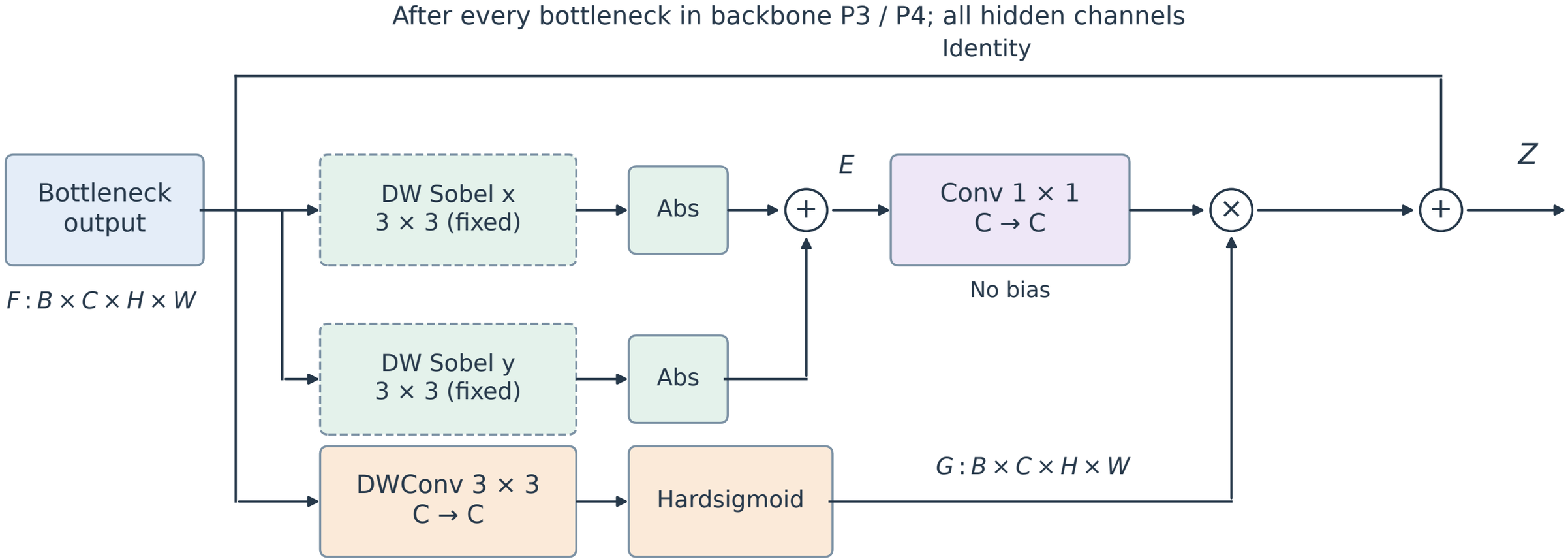
(a) SGRE | Selective Gradient Residual Enhancement

After every bottleneck in backbone P3 / P4; all hidden channels



$$Z = F + \mathrm{HS}(\mathrm{DWConv}_{3\times3}(F)) \odot \mathrm{Conv}_{1\times1}(|K_x *_{\mathrm{dw}} F| + |K_y *_{\mathrm{dw}} F|)$$